

Quantum Clock Synchronization

BU CASP Systems Lab

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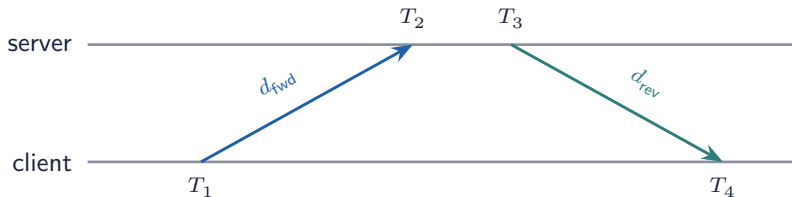
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Outline

1. Motivation
2. The QCS landscape
3. TQH-QCS
4. EB-QCS
5. Noise
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7. Evaluation
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1. **Motivation**
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Network Time Protocol



$$\delta = (T_4 - T_1) - (T_3 - T_2), \quad \Delta t = \frac{(T_2 - T_1) + (T_3 - T_4)}{2}$$

The client measures the round trip δ and **assumes** each direction took $\delta/2$. Any real difference between d_{fwd} and d_{rev} becomes an offset error of half that difference.

Spanner: Uncertainty → Latency

TrueTime

`TT.now()` returns an *interval* [earliest, latest] guaranteed to contain the true instant.

ε is half the interval width.

Commit wait

A read–write transaction assigns timestamp s , then *waits* until `TT.after( $s$ )` before releasing locks.

Expected wait $\geq 2\bar{\varepsilon}$.

Quantity	Value
Error ε	$\approx 1\text{--}7$ ms
Average error $\bar{\varepsilon}$	≈ 4 ms typically
Master poll interval	30 s
Assumed drift rate	200 $\mu\text{s/s}$

Clock uncertainty here translates to **latency**: a read–write transaction waits $2\bar{\varepsilon}$ before releasing its locks.

There is no universal time

- Simultaneity is frame-dependent, and clock rate depends on velocity and gravitational potential. **There is no clock everyone can read.**
- Time is generally coordinated on sea level, with every clock corrected into it, to reduce this disparity.

Hafele–Keating, 1971

Cesium clocks flown east and west around the world, in ns:

	Predicted	Measured
Eastward	-40 ± 23	-59 ± 10
Westward	$+275 \pm 21$	$+273 \pm 7$

The difference is that error due to relativity costs **nanoseconds**; in practice, protocols like NTP and Spanner lose **milliseconds**. That gap is caused due to an asymmetry you cannot measure, not due to the physical limit.

Can we reduce it under a different paradigm?

What QCS offers that NTP cannot

What classical timing assumes

Both directions take equal time

A timing packet can be trusted

Timing needs its own channel

Precision is bounded by how sharply you can time a pulse

What quantum resources replace it with

Quantum states tie the two events together, so the real transit time is actually measured

Quantum states cannot be copied, so interception and replay disturb the state and become visible

Synchronization is stored inside the data qubits

Entanglement can push below that bound

The quantum you need

- A qubit is a unit vector $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, with $|\alpha|^2 + |\beta|^2 = 1$.
- Measuring it returns 0 with probability $|\alpha|^2$ and destroys the superposition.
- $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ is the even split.
- The **relative phase** $e^{i\theta}$ between α and β does not affect those probabilities at all.
- A Hadamard gate H rotates phase into probability, which is how you read it.

For these protocols

The qubits are single photons.

A “measurement” is a detector click, and the hardware records *when* it clicked.

Everything reduces to times in phases and recorded timestamps.

A quantum system obeys $i \frac{d}{dt} |\psi\rangle = \mathcal{H} |\psi\rangle$, which sits as the *phase* of a qubit:

$$|\psi(t)\rangle = e^{-i\mathcal{H}t} |\psi(0)\rangle \quad (1)$$

We will take advantage of this property to allow a qubit $|\psi(t)\rangle$ to keep track of the time in its own phase.

What this buys us

$\mathcal{H}$ is really a *specification*: it fixes the relative phase as a known linear function of elapsed time.

We want a setup where reading the phase reads the time.

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Protocol	Principle	Benchmark	Limitation	Reference
★ Ticking-qubit handshake QCS	Round-trip exchange of a phase-evolving qubit; clock offset encoded in accumulated phase and extracted via interference	Quantum phase estimation determines n -bit offset using $O(n)$ qubits; coherent exchanges achieve $O(\log_2 N_c/N_c)$ uncertainty	Naive sampling requires 2^{2n} qubits; phase estimation needs extra processing qubits and inverse QFT; requires stable tick frequency	[110], [111] [113], [114]
★ Entanglement-based QCS	Pre-shared singlet state remains invariant under identical evolution; local measurements and classical communication reveal offset	Synchronization without timing-signal exchange; independent of transmission medium after entanglement distribution	Requires reliable entanglement distribution; sensitive to spatial-frame misalignment	[115]–[117]
Multiparty entanglement QCS	Multipartite entangled states (W, GHZ) shared across nodes; measurements encode and reveal local clock offsets	GHZ protocols maintain precision independent of node number; multipartite entanglement enables Heisenberg-limit scaling	W-state precision decreases with participants; requires global entanglement distribution and joint measurements	[118]–[123]
Time-of-arrival QCS	Clock offset inferred from photon arrival-time correlations; frequency entanglement and number squeezing reduce timing uncertainty	Classical scaling: $\sqrt{M}$ pulses and $\sqrt{N}$ photons; entanglement and squeezing yield $\sqrt{MN}$ improvement	Highly sensitive to photon loss; multi-pulse entanglement difficult to generate over long links	[1]
Conveyor-belt QCS	Bidirectional exchange of modulated optical pulses; offset derived from returned interference pattern	Insensitive to dispersion; frequency-entangled pulses allow quantum dispersion cancellation	Requires equal forward/backward transit times and stable clocks during signal round-trip	[51], [125]
HOM synchronization	Balances HOM interferometer using reflected entangled photons; correlated arrivals reveal clock offset	Distance-independent synchronization; suitable for optical and space links	Requires interferometer balancing and stationary source configuration	[126]
Time-offset QCS	Cross-correlation of entangled-photon detection timestamps estimates time offset and frequency drift	FFT peak detection enables sub-picosecond precision and ppm-level drift estimation	Accuracy limited by detector jitter, noise, and coincidence statistics	[71], [106] [127], [128]
Two-way time-transfer QCS	Signal photons exchanged from frequency-entangled pairs while idlers remain local; transit delays cancel	Quantum correlations enable femtosecond-level synchronization precision	Requires SPDC sources, accurate time tagging, and broadband biphoton spectra	[129], [130]
Space-to-ground QCS	Satellite-borne quantum resources distribute timing globally via optical links	Sub-nanosecond global timing distribution and high-precision satellite synchronization	Requires space-qualified quantum hardware and robust channel models	[73]–[75]

Two categories of protocols

■ State preparation and estimation

The time is stored in a phase you prepare, let evolve, and measure.

■ Interference and arrival times

The time is stored in how photons interfere.

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Background: the phase serves as the clock

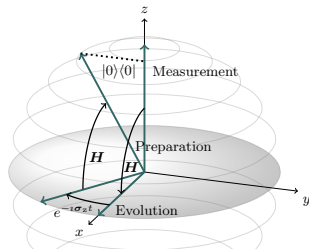
Left alone under the Hamiltonian $\mathcal{H} = \omega\sigma_z$, the state $|+\rangle$ never changes its measurement statistics. It only turns its phase, at a fixed rate ω :

$$e^{i\omega t \sigma_z} |+\rangle = \frac{1}{\sqrt{2}} (e^{i\omega t} |0\rangle + e^{-i\omega t} |1\rangle)$$

Apply H and that hidden phase becomes visible:

$$\cos(\omega t) |0\rangle + i \sin(\omega t) |1\rangle, \quad \text{Pr}[0] = \cos^2(\omega t)$$

That rotation is the tick. A qubit in transit is a clock that keeps running while it travels.

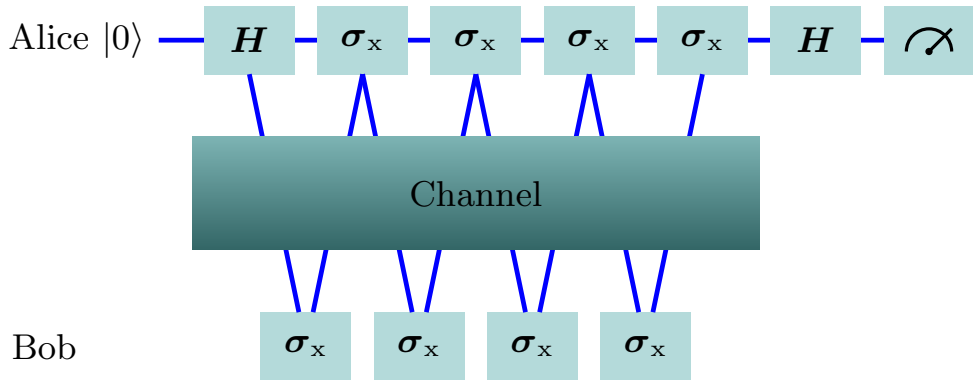


Ticking-qubit handshake

- A single qubit is the message *and* the clock.
- Its phase advances at a known tick rate ω while in transit.
- Alice and Bob bounce it back and forth, each applying σ_x .
- The accumulated phase encodes Δt , and Alice reads it out.

Structurally this is pretty similar to an NTP round trip. The difference is that the timing is being physically carried by the object being sent.

TQH-QCS



Alice transmission

Write coordinate time as t , with the two clocks reading $C_A(t) = t + \delta_A$ and $C_B(t) = t + \delta_B$. The unknown is $\Delta t \equiv \delta_B - \delta_A$, and the true transits T_{AB} , T_{BA} are **never assumed equal**.

Alice emits $|\psi\rangle$ at t_1 , her clock reading $t_A = t_1 + \delta_A$, and announces that number. In transit the qubit evolves as $|\psi(t)\rangle = e^{-i\mathcal{H}t} |\psi\rangle$, so with $\mathcal{H} = \omega\sigma_z$ it arrives carrying $e^{-i\omega T_{AB}\sigma_z} |\psi\rangle$, while Bob's clock reads $t_B = t_1 + T_{AB} + \delta_B$:

$$t_B - t_A = T_{AB} + \Delta t \quad (4)$$

He applies $\sigma_x e^{i\omega(t_B - t_A)\sigma_z}$ on receipt, and the transit drops out:

$$\sigma_x e^{i\omega(T_{AB} + \Delta t)\sigma_z} e^{-i\omega T_{AB}\sigma_z} |\psi\rangle = \sigma_x e^{i\omega\Delta t\sigma_z} |\psi\rangle \quad (5)$$

The phase records the **actual** Δt value, which is exactly what we need.

Bob transmission

Bob sends it back and it accumulates $e^{-i\omega T_{BA}\sigma_z}$. Alice receives at $t_{A'}$, Bob announced $t_{B'}$, and the two clocks now enter in the *opposite* order:

$$t_{A'} - t_{B'} = T_{BA} - \Delta t \quad (6)$$

Alice applies $\sigma_x e^{i\omega(t_{A'} - t_{B'})\sigma_z}$, and $\sigma_x e^{i\theta\sigma_z} = e^{-i\theta\sigma_z} \sigma_x$ finishes it:

$$\begin{aligned} \sigma_x e^{i\omega(T_{BA} - \Delta t)\sigma_z} e^{-i\omega T_{BA}\sigma_z} \sigma_x e^{i\omega\Delta t\sigma_z} |\psi\rangle \\ = \sigma_x e^{-i\omega\Delta t\sigma_z} \sigma_x e^{i\omega\Delta t\sigma_z} |\psi\rangle = e^{2i\omega\Delta t\sigma_z} |\psi\rangle \end{aligned} \quad (7)$$

Without the σ_x flips the return leg would contribute $e^{-i\omega\Delta t\sigma_z}$ and undo the forward leg exactly. Conjugation reverses the sign, so the two legs **add**.

Reading the phase out

Δt is generated by σ_z , so $|\psi\rangle$ cannot be $|0\rangle$ or $|1\rangle$, or Δt would factor out and be unmeasurable. Take $|\psi\rangle = |+\rangle$:

$$e^{2i\omega\Delta t\sigma_z} |\psi\rangle = \frac{1}{\sqrt{2}} (e^{2i\omega\Delta t} |0\rangle + e^{-2i\omega\Delta t} |1\rangle) \quad (8)$$

where $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$.

Measuring (8) directly gives nothing, since the outcome probabilities are the squared norms of those amplitudes. Applying H interferes them:

$$\cos(2\omega\Delta t) |0\rangle + i \sin(2\omega\Delta t) |1\rangle$$

Alice sees $|0\rangle$ with probability $\cos^2(2\omega\Delta t)$. Repeating the exchange estimates that probability, and with it Δt .

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Background: entanglement

- Two qubits can share a joint state that is not a pair of individual states.
- The **singlet** $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ is the canonical one: neither qubit has a definite value, yet the two are guaranteed opposite.
- Measuring one fixes the other's outcome immediately, at any separation, with nothing sent between them.
- This transmits no information, but it does give both parties a **shared reference** each can measure against locally.

A pre-shared entangled pair is a resource you distribute once and spend later.

Entanglement-based synchronization

- Alice and Bob share a singlet *before* they need the time.
- The singlet does not evolve: identical local evolution cancels out.
- No timing signal is ever transmitted.
- So the protocol is insensitive to the transmission medium.

There is no channel to model and no delay to estimate, because at synchronization time nothing crosses the network, i.e. there is no online quantum communication.

The shared state

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|01\rangle_{AB} - |10\rangle_{AB}) \quad (9)$$

Equivalently, in the Hadamard basis,

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|+-\rangle_{AB} - |-+\rangle_{AB}) \quad (10)$$

Under $\mathcal{H} = \omega(\sigma_z^A + \sigma_z^B)$ both $|01\rangle$ and $|10\rangle$ have energy zero, so the singlet is an **eigenstate of the joint evolution** and does not move at all. Nothing is ticking yet.

Measuring is what **starts the clock**: it leaves Bob holding a state whose phase grows with the time elapsed since Alice acted.

Where the offset enters

Same clock model as TQH-QCS: $C_A(t) = t + \delta_A$, $C_B(t) = t + \delta_B$, unknown $\Delta t = \delta_B - \delta_A$.

At coordinate time t_1 , her clock reading $t_A = t_1 + \delta_A$, Alice measures in $\{|\pm\rangle\}$ and collapses Bob's qubit to $|\mp\rangle$. Bob measures at t_2 , his clock reading $t_B = t_2 + \delta_B$, on the state

$$e^{-i\omega t \sigma_z} |\mp\rangle, \quad t \equiv t_2 - t_1 \quad (11)$$

The phase gives him t ; once Alice announces t_A , the clocks give him $t_B - t_A$:

$$t_B - t_A = t + \Delta t \quad \implies \quad \Delta t = (t_B - t_A) - t \quad (12)$$

Just like TQH-QCS, we can extract the difference Δt from the clock times t_A, t_B and the time t in the phase.

Bob applies H and measures. The two branches are not the same:

$$|-\rangle_A \Rightarrow |+\rangle_B : \quad \Pr[0] = \cos^2 \omega t \quad (13)$$

$$|+\rangle_A \Rightarrow |-\rangle_B : \quad \Pr[0] = \sin^2 \omega t \quad (14)$$

so Alice must send her *outcome*, not just her timestamp. Repeating over many pairs estimates the probability, and with it t .

Space has to be synchronized too

The protocol assumes both parties agree on which direction is z . If their frames differ by polar angle θ and azimuth ϕ , the measurement probabilities are distorted. Alice also measures in the computational basis; Bob post-selects and evaluates

$$\langle - | \sigma_z | - \rangle_B = \sin \theta \cos \omega t \quad (15)$$

$$\langle 0 | \sigma_y | 0 \rangle_B = \sin \theta \sin \phi \quad (16)$$

or, post-selecting instead on $|1\rangle_B$ and $|0\rangle_B$, the simpler pair

$$\langle 1 | \sigma_z | 1 \rangle_B = \cos \theta \quad (17)$$

$$\langle 0 | \sigma_x | 0 \rangle_B = \sin \theta \cos \phi \quad (18)$$

Bob recovers θ and ϕ , rotates his frame to match Alice's, and only then runs the clock protocol.

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Each protocol has its own dominant noise

Protocol	What carries the time	What destroys it
TQH-QCS	phase of a qubit in repeated transit	decoherence
EB-QCS	relative phase of a stored singlet	phase error

TQH-QCS: decoherence

Channel	Single-qubit transport	n entangled qubits
Bit flip	$\frac{1}{2}$	$\frac{1}{n}(1-p)^n$
Phase flip	$\frac{1}{2}(1-2p)^2$	$\frac{1}{n} (1-2p)^n $
Amplitude damping	$\frac{1}{2}(1-p)$	$\frac{1}{n}(1-p)^{n/2}$

- Bit flips barely matter, because the time is stored in the phase.
- Phase flips are quadratic, and the repeated-exchange variant is **exponential** in p : precision diverges as $p \rightarrow 0.5$.
- Every extra round trip crosses the channel again and accumulates more error.

A phase η slipped into the shared singlet gives

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|0\rangle_A |1\rangle_B - e^{i\omega\eta} |1\rangle_A |0\rangle_B) \quad (19)$$

and the protocol reports $t - \eta$. This leads to an **incorrect** result.

- Purification would repair the state, but purification itself requires synchronous operations and a shared basis convention. That is circular.
- Error correction would protect the state, but any code stable against phase errors also suppresses the phase evolution that carries the time.
- Under complete dephasing the timing information is gone outright, and no separable or entangled strategy recovers it.

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Possible attack vectors

Threat	Hits	Countermeasure	Feasibility
Entanglement hijacking	EB-QCS	Bell and polarization checks	very low
Intercept-and-resend	all	QKD authentication, QBER monitoring	impossible ideally
Symmetric delay	one-way	sampling, indistinguishability checks	low
Asymmetric delay	two-way	Franson, asymmetry monitoring	extremely difficult
Photon-number splitting	weak pulses	decoy states, QBER statistics	partial
Time-stamp spoofing	classical channel	authenticated encryption, replay protection	moderate
Detector side-channel	the receiver	power limiting, monitoring, calibration	moderate
Channel delay manipulation	the fiber	two-way transfer, delay calibration	moderate
Denial of service	availability	monitoring, redundancy, path diversity	high

The quantum-specific attacks are the hard ones. Everything below the rule is an ordinary systems problem, i.e. QCS doesn't add any likely quantum-specific attacks.

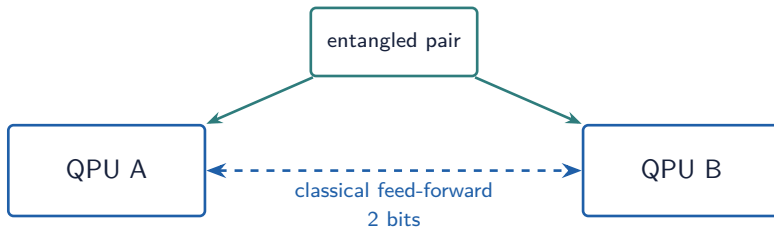
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Protocol	Implementation	Performance
TQH-QCS	3-qubit NMR, 2004	>100 ps (fidelity 75–91%)
EB-QCS	4-qubit NMR, 2018	29- μ s synchronization delay
NTP	deployed, global	1–10 ms over WAN
White Rabbit	deployed, CERN	0.517 ± 0.119 ns over 16 km fiber

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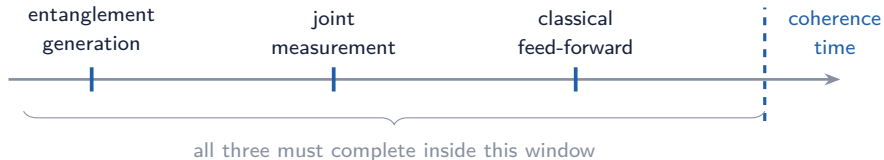
What sets ε	Today	What QCS changes
Oscillator drift between polls	$200 \mu\text{s/s} \times 30 \text{ s}$	nothing — already cheap
Round trip to the time master	$\sim 1 \text{ ms}$	attacks this term
Path asymmetry	unmeasurable	two-way cancels it
Trust in the time source	assumed	tamper-evident

Another application: Distributed quantum computing



- One quantum computer is hard to scale, so link several into one machine.
- A gate across two QPUs is run by *teleporting* state over a shared pair.
- That costs one entangled pair, a joint measurement, and two classical bits.

Why the clock is load-bearing



- This is a hard physical time limit correlated with the state of current quantum hardware.
- Miss it and the computation fails and must be abandoned.
- Meeting it needs a **shared temporal reference** across nodes, i.e. QCS.

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